AR Advertisement Detection & Replacement

Reducing visual clutter through real-time AR advertisement blurring and replacement

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Motivation

1

Visual overload

City centers are saturated with advertising, competing for pedestrians' attention.

2

Cognitive load

Constant advertising exposure increases stress and mental effort while walking.

3

AR as a filter

Instead of adding more content, AR can be used to remove or soften visual noise.

Research question: Does automatically detecting and blurring/replacing advertisements improve usability, reduce workload, and increase immersion - without lowering trust?

My task

Concept	Designed the AR advertisement detection & replacement concept, including an initial navigation feature that was later removed for safety/testability reasons.
Implementation	Built the Unity prototype: detects advertisement surfaces and applies a real-time blur or replaces them with a relaxing video.
Study design	Designed a within-subject user study with 3 conditions, 4 validated questionnaires (SUS, NASA-TLX, Trust, ARI), and a behavioral distraction task.
Data collection	Set up and ran the study in LimeSurvey with 18 participants across all three conditions.
Analysis	Cleaned, merged, and statistically analyzed the data (Shapiro-Wilk, RM-ANOVA / Friedman, post-hoc tests).

Concept overview

Capture

Camera view of the pedestrian's real-world surroundings

→

Detect

Advertisement surfaces (billboards, screens) are recognized in real time

→

Decide

System applies one of two treatments to the detected surface

Condition B - Blur

The detected advertisement is blurred in real time, reducing its visual salience while keeping spatial context intact.

Condition C - Replacement

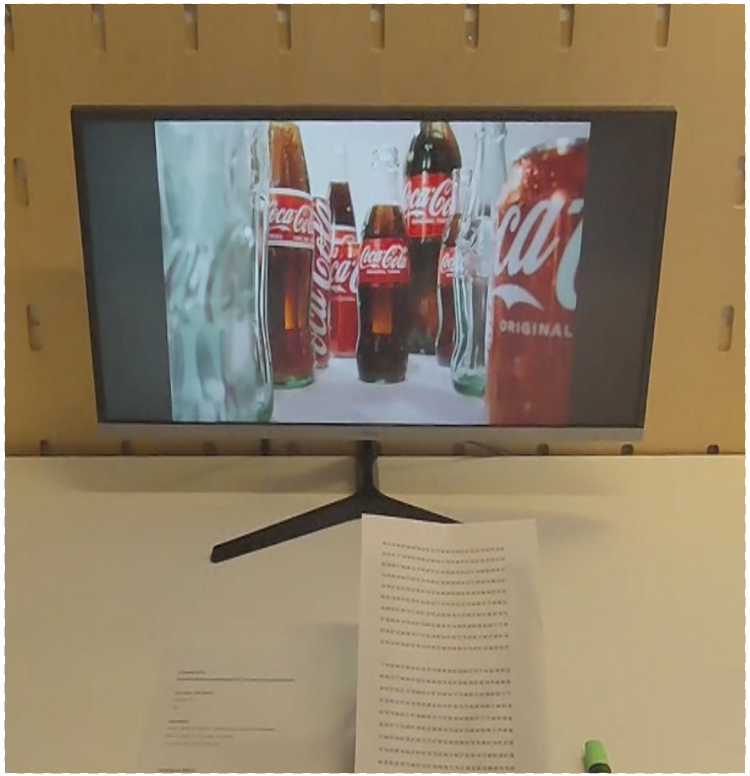
The advertisement is replaced with a short relaxing video, turning a distracting surface into a calming visual element.

Note: an earlier version also included a pedestrian navigation overlay. It was removed, as testing it safely in real outdoor traffic was not feasible within this project.

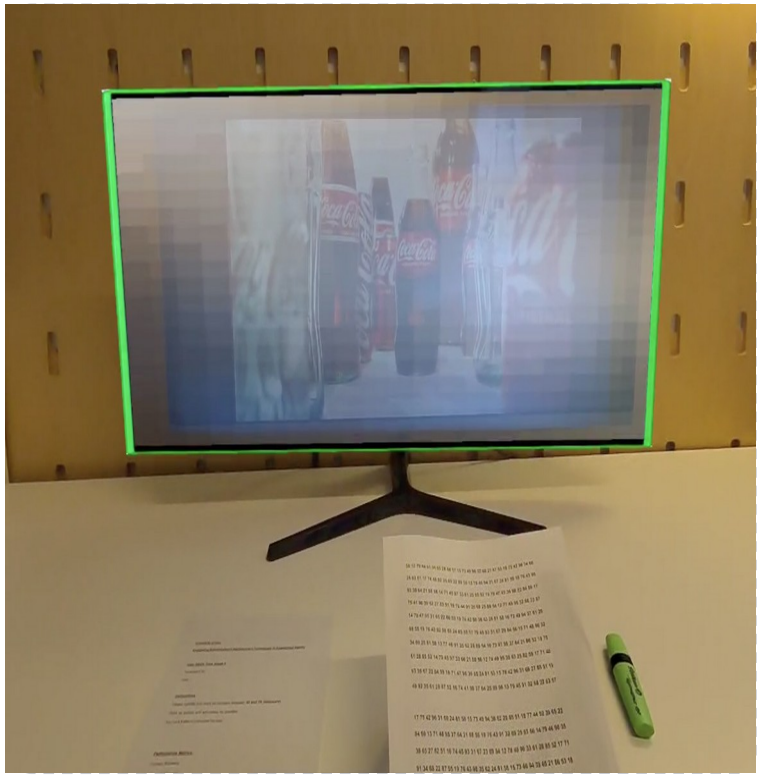
Technical implementation

Platform	Meta Quest 3 (video passthrough AR headset)
Engine	Unity, C# (based on Meta's Passthrough Camera API samples)
Detection	YOLOv9 model, converted to Unity Sentis, running on-device (CPU backend) to detect advertisement surfaces in real time
Stability	Detections are filtered across multiple frames (median-based) before the overlay position is locked in place
Treatment	Detected regions are either blurred (shader-based) or overlaid with a video texture

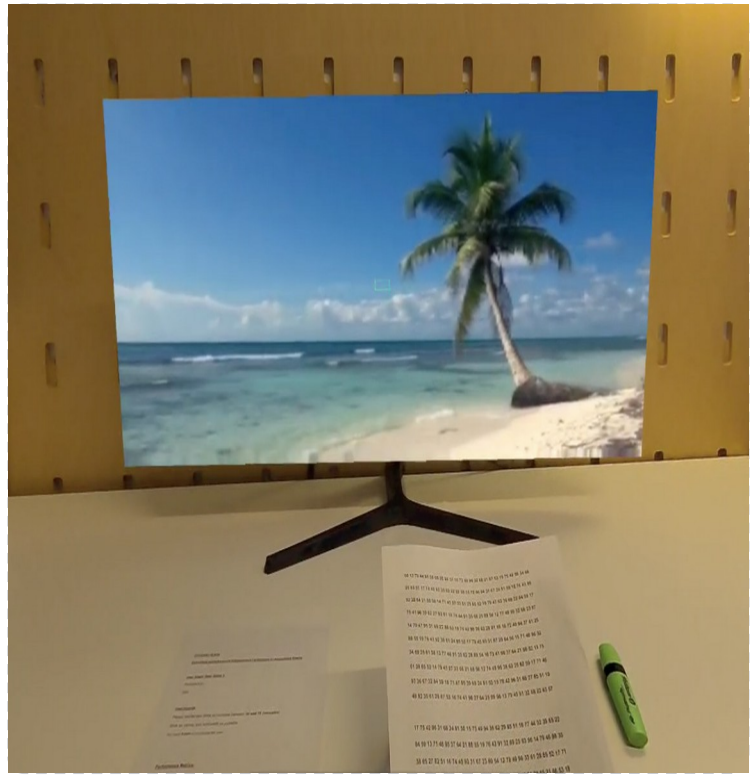
Prototype



A - Original advertisement



B - Blurred advertisement



C - Advertisement replaced

Study design

18	3	4	1
Participants	Conditions (within-subject)	Questionnaires per condition	Behavioral distraction task

Conditions

A Original advertisements
unmodified

B Blurred advertisements
diminished

C Advertisement replacement
altered

Based on mediated reality (Mann; MIRAGE)

Measures

SUS: perceived usability

NASA-TLX (raw): perceived workload

Trust in Automation: trust

ARI: immersion

Distraction task: error rate (objective)

Sample

18 participants, within-subject

Guideline: 36 (HCI Toolkit)

Convenience sample

Smaller sample limits statistical power

Sources: Mann & Barfield (2003); MIRAGE (CHI 2026); Schwind et al. (2023), HCI User Studies Toolkit.

From navigation to a paper-based task

Used in the study: paper task

User Study Task Sheet

Instructions

Please identify and mark all numbers between **30 and 70 (inclusive).**

Work as quickly and accurately as possible.

You have **3 min** to complete the task.

38 12 74 95 21 44 87 63 19 55 32 78 91 46 29 67 15 83 50 24 71 96 35 11 89

62 81 45 27 93 16 58 39 72 84 25 61 14 98 47 75 33 52 18 97 41 68 22 86 54

23 59 88 17 64 31 79 53 26 92 48 13 85 69 34 76 94 51 28 65 12 82 43 77 36

71 46 29 83 55 18 96 42 67 24 81 59 35 73 91 16 88 49 22 64 37 75 51 94 28

Excerpt of the number grid

Original idea: navigation



Challenge: Safety risks when testing navigation

Solution: Paper-based task, completed with the headset on

Distraction measure: Error rate (objective)

Note: Project partner dropped out early; remaining work completed individually.

Statistical approach

A = Original Advertisements · B = Blurred Advertisements · C = Advertisement Replacement

1. Compute scores — SUS, NASA-TLX, Trust, ARI, error rate - one score per participant per condition

2. Shapiro-Wilk test — Check normality of each score, separately for A, B and C

normal in all 3 conditions

One-way repeated-measures ANOVA (Trust)

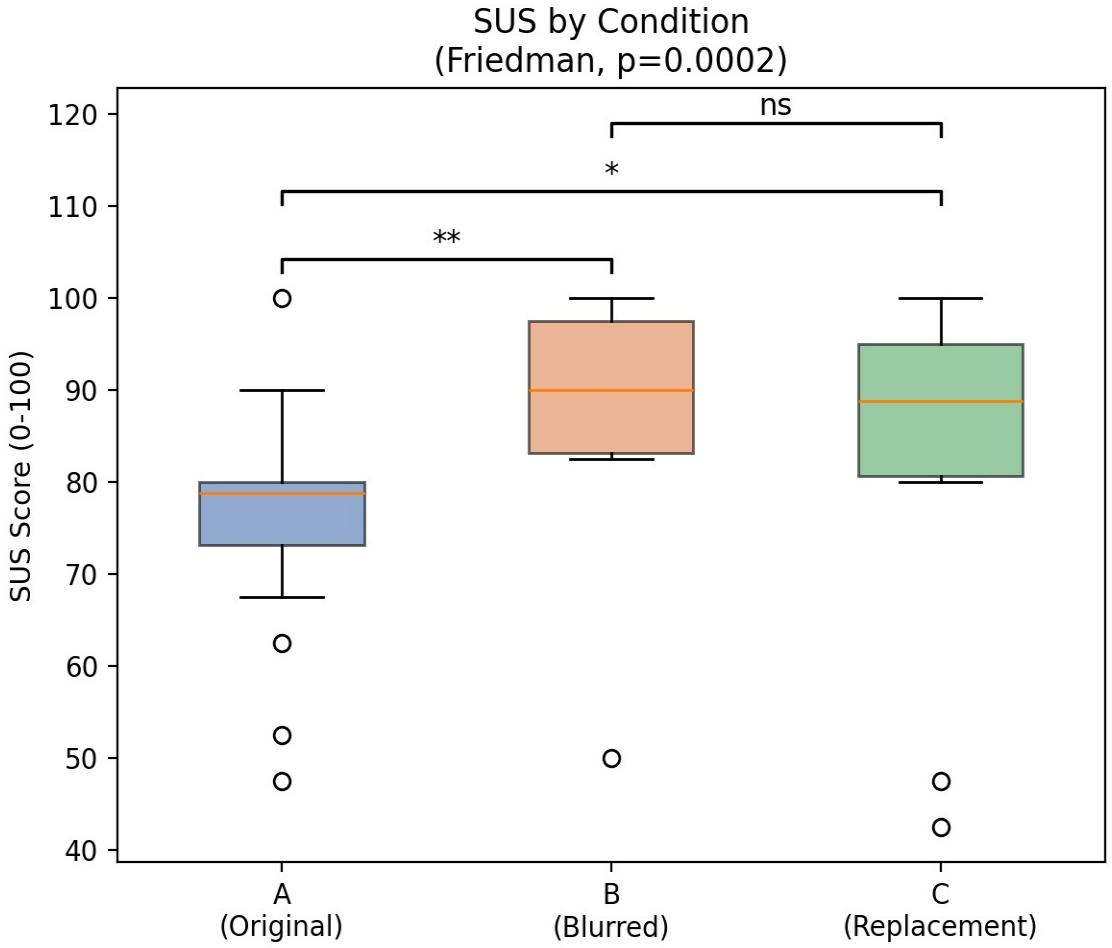
Post-hoc: paired t-tests + Holm correction

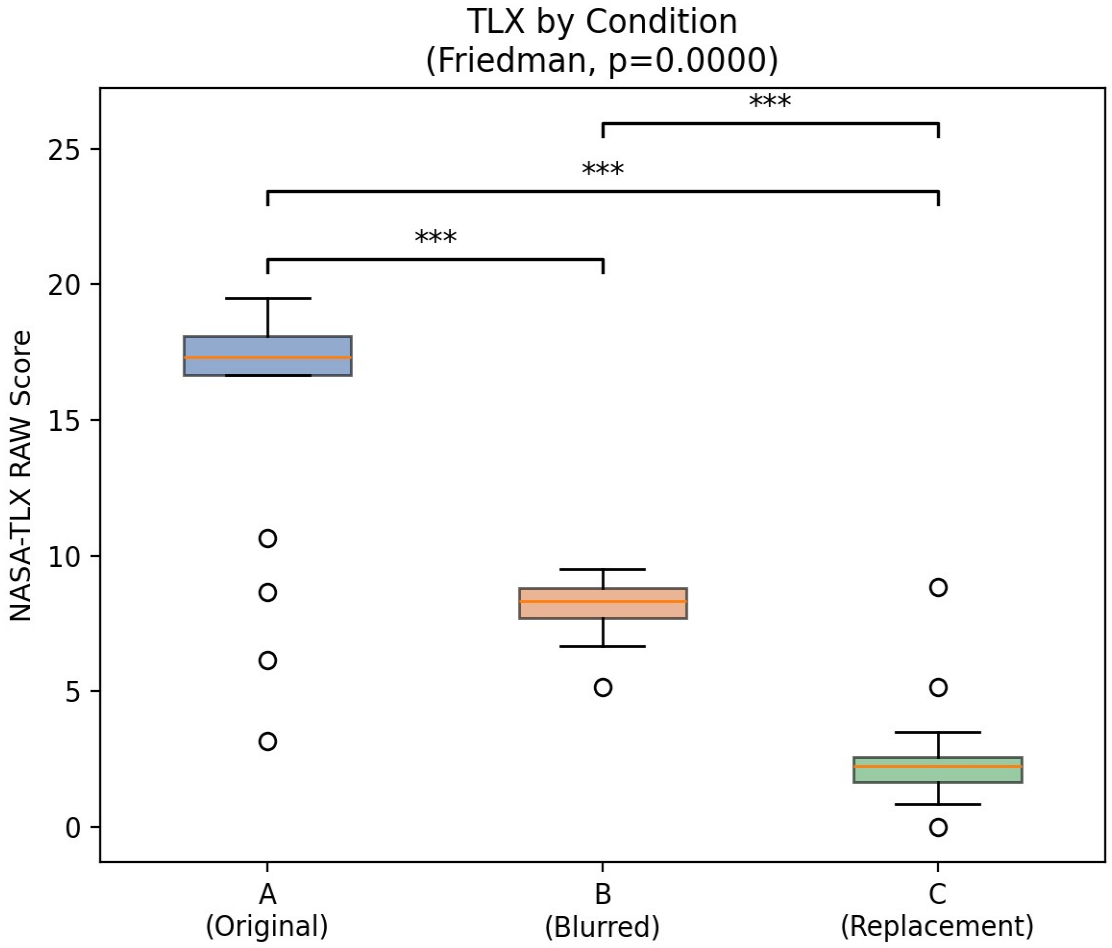
not normal in at least one

Friedman test (SUS, TLX, ARI, error rate)

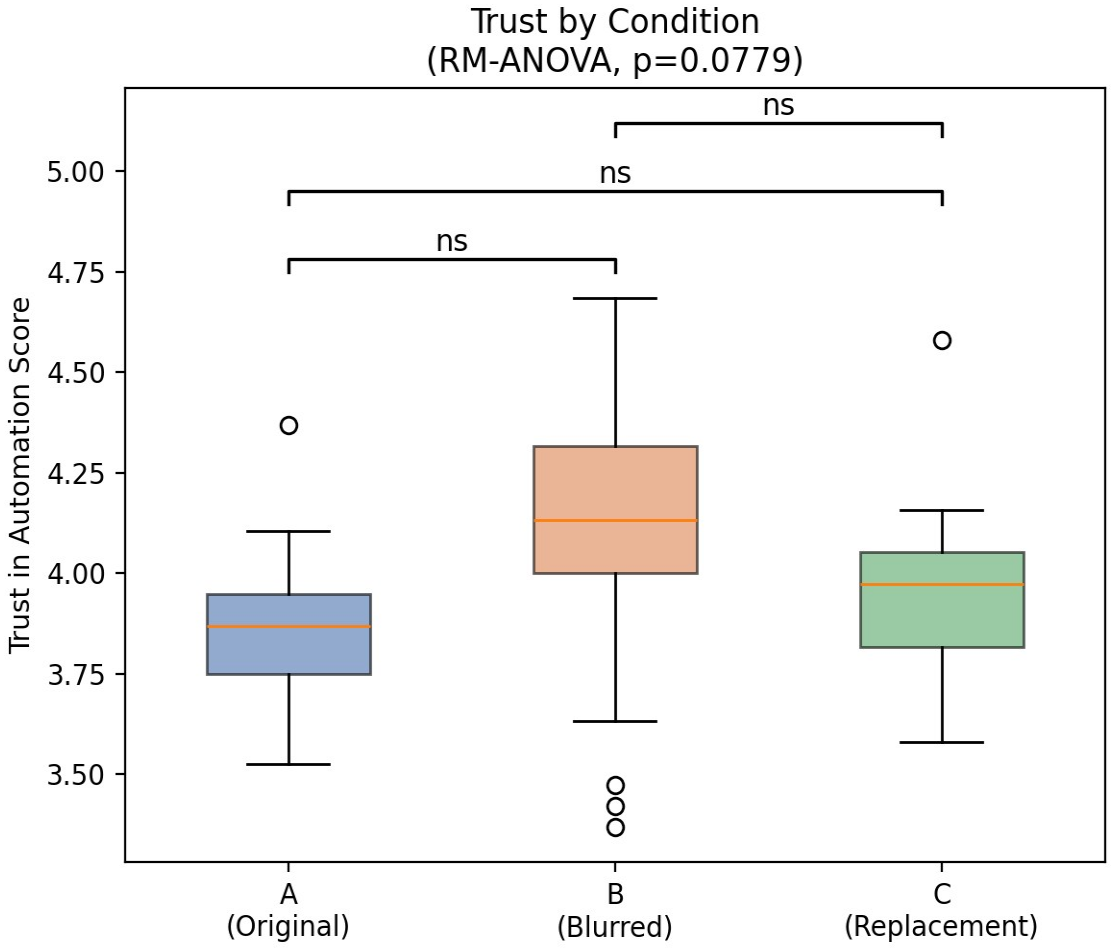
Post-hoc: Wilcoxon signed-rank + Holm correction

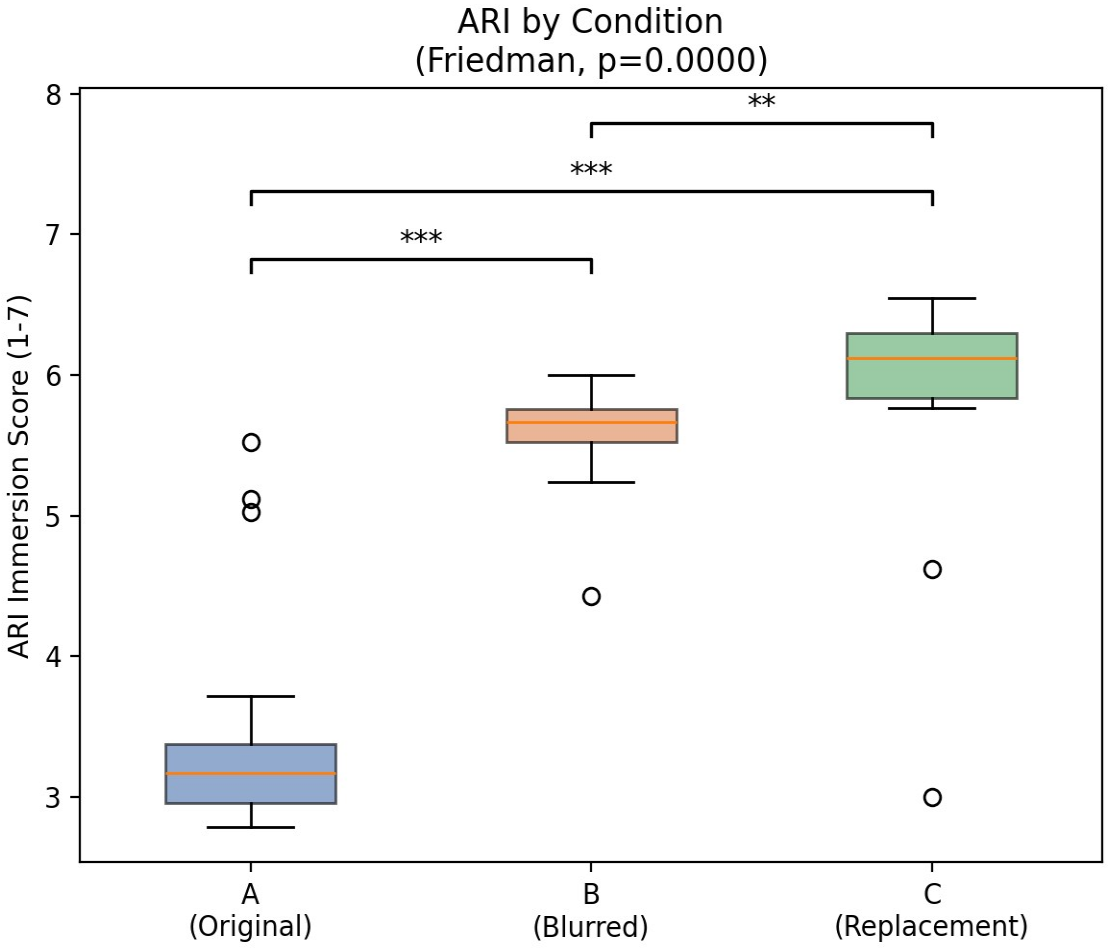
Results - usability & workload





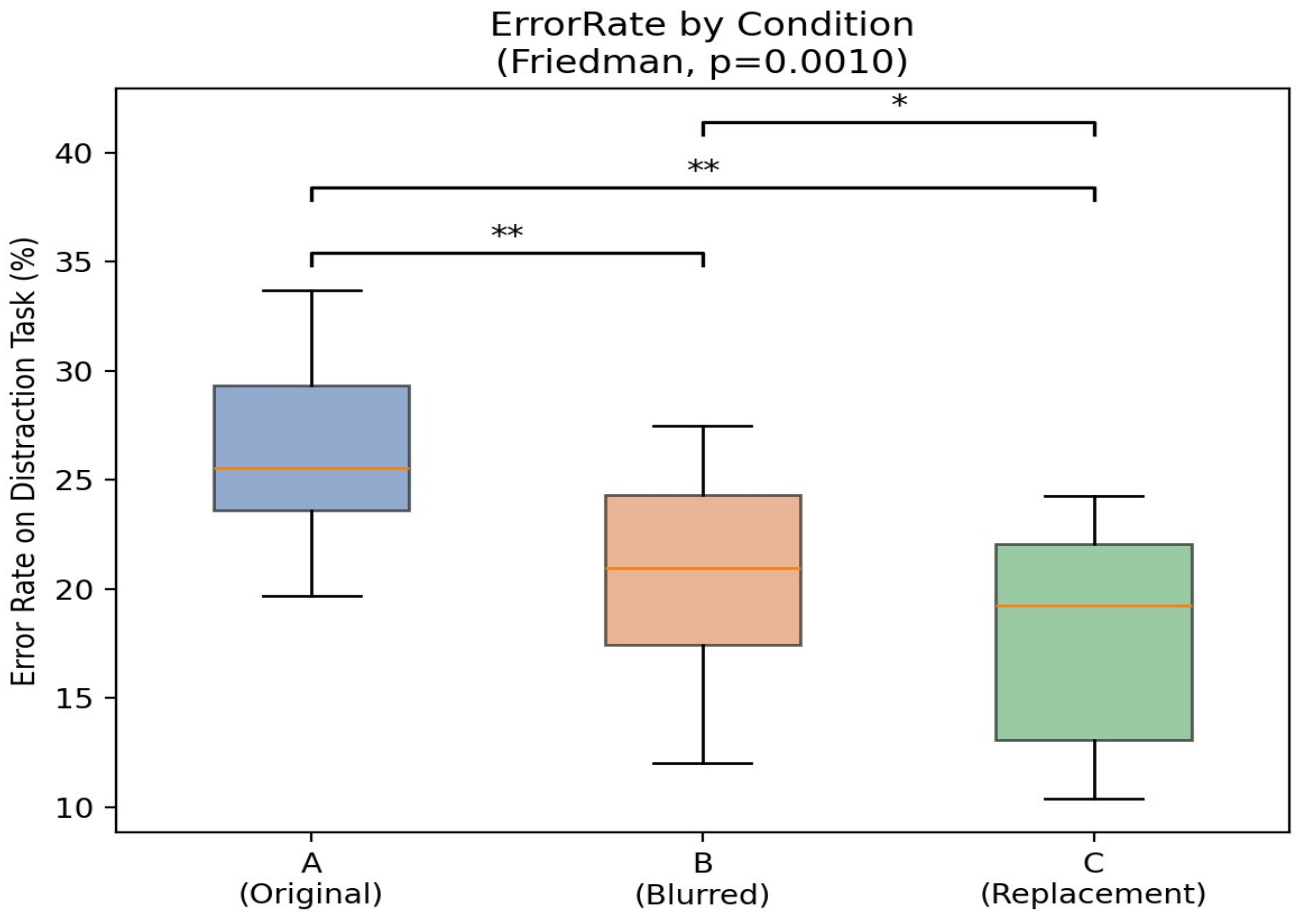
Results - trust & immersion





Results - distraction task

Objective behavioral measure: error rate on the number-finding task, done once per condition.



Discussion & conclusion

A = Original Advertisements · B = Blurred Advertisements · C = Advertisement Replacement

Usability ↑	Blurring or replacing ads (B, C) rated significantly more usable than the original view (A).
Workload ↓	Mental workload decreased clearly and significantly from A to B to C.
Trust →	No significant difference in trust - modifying ads did not make the system feel less trustworthy.
Immersion ↑	Immersion was significantly higher in B and C than in A.
Distraction ↓	Error rate on the distraction task also decreased significantly from A to B to C, confirming the questionnaire results with an objective measure.
Limitation	Small convenience sample (N = 18, guideline: 36) limits statistical power and generalizability.

Removing or replacing advertisements in AR makes the experience more usable, less demanding, and more immersive - without costing user trust.

References

Mann, S., & Barfield, W. (2003). Introduction to mediated reality. Int. J. Human-Computer Interaction, 15(2), 205-208.

MIRAGE: Enabling Real-Time Automotive Mediated Reality. CHI 2026. doi:10.1145/3772318.3791195

Schwind, V., Resch, S., & Sehrt, J. (2023). The HCI User Studies Toolkit. CHI EA '23. doi:10.1145/3544549.3585890

Brooke, J. (1996). SUS: A quick and dirty usability scale. In Usability Evaluation in Industry. Taylor & Francis.

Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX. Advances in Psychology, 52, 139-183.

Körber, M. (2019). Theoretical considerations and development of a questionnaire to measure trust in automation. IEA 2018 Proceedings.

Georgiou, Y., & Kyza, E. A. (2017). The development and validation of the ARI questionnaire. Int. J. Human-Computer Studies, 98, 24-37.

Wang, C.-Y., Yeh, I.-H., & Liao, H.-Y. M. (2024). YOLOv9: Learning what you want to learn using programmable gradient information. ECCV 2024.

Meta Platforms Technologies. Unity-PassthroughCameraApiSamples (GitHub repository).

Thank you

Questions?